PART 06 · CHAPTER 26

Designing a complete trading system

A signal is one component of a much larger decision process.

Learner edition: attempt first; continuation reference starts on a new page where supplied. Worked solutions and quiz answers are omitted.

advanced About 15 min of prose · practice takes additional time

Draft manuscript · source notes below

> THE IDEA TO KEEP
>
> A complete system specifies data, eligibility, ranking, sizing, fills, exits, and accounting so precisely that two people processing the same inputs produce the same ledger.

By the end of this chapter

- Screen, rank with a deterministic tie-break, and size simultaneous candidates as the smallest of the risk, notional, and cash allowances.
- Process fills, partial fills, nonfills, halts, gaps, same-bar ambiguity, and time exits in a declared order.
- Keep a ledger in which cash + holdings = equity at every close, with reserved cash and costs, and report net R on each trade's frozen 1R.
- Distinguish definitions, conventions, and arbitrary teaching settings, and explain why any rule change, including a regime filter, is a new version.

Builds on: Breakouts, pullbacks & failed moves · Events and opening gaps · Market microstructure: from a chart to an actual trade · Risk & position sizing · From chart idea to trade plan · A journal that can teach you something · Evidence before conviction

First pass: reconcile the S4–S5 accounting replay

This worksheet begins from the supplied S3-close state. Cash is $54,714.02, the five filled holdings are listed below, and no entry orders are working. Each prior entry incurred

$2.50; an exit incurs another $2.50. Use filled shares, not earlier order quantities, to calculate each position's frozen R.

This is an accounting replay, not a forecast or an unseen-chart learning test. The S4/S5 bars and screen facts are exercise inputs, released for this replay. Process S4 and its decision close before using S5. The resulting exits, cash, equity and full reference ledger stay behind the continuation reveal. All stocks and records are fictional.

The three unchanged v1.0 rule tables are restated here so the worksheet is self-contained. For each session, record exits in priority order, cash after fills, every surviving close mark, equity, occupied/free slots and any new order. Then give Fir's and Birch's net trade dollars and net R. Start Fir's trade P&L from its $36.00 entry, even though the replay starts from a later close mark.

Fir's S5 bar is not supplied. If your S4 processing leaves it held, flag that missing input rather than inventing a mark. For D4/D5, the supplied screen facts cover the unheld candidates; no omitted stock supplies an additional signal. Freeze your working below before opening the complete continuation.

Starting position state at S3 close (fictional USD; Fir mark stale)

Position	Filled shares	Entry session / fill ($)	Stop / target ($)	Frozen 1R ($)	Last close ($)
Fir	250	S−2 / 36.00	34.00 / 40.00	500.00	35.10 (stale: S3 has no bar)
Alder	250	S1 / 32.10	30.10 / 36.10	500.00	33.90
Birch	118	S1 / 72.20	70.20 / 76.20	236.00	73.10
Dogwood	158	S2 / 62.80	60.80 / 66.80	316.00	62.20
Cedar	234	S3 / 42.72	40.72 / 46.72	468.00	43.10

Atlas-20 v1.0 rule sheet, part 1: data, universe, and eligibility

Rule	Atlas-20 v1.0 setting	Kind
Scope	Long-only fictional U.S.-listed common stocks in one U.S.-dollar account; no leverage, short sales, or derivatives.	Scope choice

Rule	Atlas-20 v1.0 setting	Kind
Universe	Point-in-time member list as of each decision date. Scenario members throughout: Alder, Birch, Cedar, Dogwood, Elm, Fir, Ginkgo, Hazel.	Convention
Data and clock	Completed regular-session daily OHLC and share volume; split adjustment uses only actions effective by the decision date. Decisions are computed after the close; orders are submitted before the next open.	Convention
Minimum history	21 consecutive valid bars: sessions t−20 through t.	Teaching setting
Missing data	Any absent or invalid bar in t−20…t: no signal is evaluated that day. No interpolation or carried-forward bars.	Convention
Already held	A stock with an open position or a working order is not screened; v1.0 never adds to a position.	Teaching setting
Signal	Close at t strictly above the highest high of sessions t−20…t−1.	Definition
Price band	$20.00 ≤ decision close ≤ $100.00, so the fixed $2 stop is 2%–10% of price.	Teaching setting
Liquidity	Mean of close × volume over sessions t−19…t ≥ $2,000,000 (past data only).	Teaching setting
Earnings	No scheduled earnings in sessions t+1…t+10 (the planned ten-session holding window; a halt can delay exit), per the calendar snapshot at the decision close.	Teaching setting
Regime filter	None in v1.0.	Scope choice
Check order	Complete record → signal → price band → liquidity → earnings. The first failure is recorded.	Convention

Atlas-20 v1.0 rule sheet, part 2: ranking, slots, sizing, and costs

Rule	Atlas-20 v1.0 setting	Kind
Ranking value	Breakout margin = 100 × (close − prior-20 high) ÷ prior-20 high, rounded to 0.01 percentage point; larger ranks first.	Teaching setting
Tie-break	Alphabetical by company name.	Convention
Slots	At most 5 = open positions (halted ones included) + orders placed at this close.	Teaching setting
Sizing equity E	Cash + shares × close for every holding, after the session’s exits and fills; a halted holding uses its last valid close.	Definition
Limit price L	1.01 × signal close, rounded down to the cent.	Teaching setting
Risk allowance	floor(0.5% × E ÷ $2.00) shares.	Teaching setting
Notional allowance	floor(10% × E ÷ L) shares.	Teaching setting
Cash allowance	floor((cash − cash already reserved at this close − $2.50) ÷ L) shares.	Definition
Quantity	Smallest of the three allowances; fewer than 1 share means no order.	Definition
Cash reservation	Shares × L + $2.50, held from order placement until the order fills or expires.	Convention
R	1R = filled shares × $2.00, frozen at the fill. Net R = net P&L ÷ 1R. Costs are shown separately.	Book contract
Costs	$2.50 per executed fill (entry, exit, or partial entry), so $5.00 per round trip. No separate spread or slippage charge beyond the fill rules.	Teaching setting
Cash interest and settlement	Cash earns zero; settlement timing is ignored.	Convention

Atlas-20 v1.0 rule sheet, part 3: orders, fills, exits, and review

Rule	Atlas-20 v1.0 setting	Kind
Entry order	Buy day-limit order at L for the next session only; any unfilled remainder cancels at that session’s close.	Convention

Rule	Atlas-20 v1.0 setting	Kind
Entry fill	Open ≤ L: fill at the open. Otherwise, low strictly below L: fill at L. A low exactly at L is a touch, not a fill.	Convention (conservative)
Partial fill	Filled shares ≤ floor(1% × entry-session volume). Keep the filled shares, cancel the rest, never top up.	Teaching setting
Rejected or unfilled candidates	Not re-offered. A stock needs a new signal at a later close.	Convention
Stop	Sell stop at fill − $2.00; triggered when low ≤ stop.	Teaching setting
Target	Sell limit at fill + $4.00; fills only when high is strictly above the target.	Teaching setting
Time exit	At the close of holding session 10 (entry session = 1; exchange sessions counted, including halted ones).	Teaching setting
Exit priority (held from an earlier session)	1. Open ≤ stop: exit at open. 2. Open ≥ target: exit at open. 3. Low ≤ stop and high > target: stop assumed first. 4. Low ≤ stop: exit at stop. 5. High > target: exit at target. 6. Holding session 10: exit at close. After opening checks 1–2, an overdue time deadline exits at the open before steps 3–5; a scheduled session-10 close uses step 6.	Convention (conservative)
Entry bar	Fill at the open: steps 3–5 apply to the rest of that bar. Fill at L intraday: the stop applies if low ≤ stop; a target is not credited.	Convention (conservative)
Halt or missing bar while held	No exit is possible. Mark at the last valid close, flagged stale; the position keeps its slot. The first new bar starts at step 1.	Convention
Processing order	Batch accounting: held-position exits and reserved entries, then close marks and screening. Entry quantities are fixed at the previous close and fully cash-reserved; no same-session exit funds a larger order. Within-session event rows are not a timestamped tape.	Convention

Rule	Atlas-20 v1.0 setting	Kind
Review triggers	Operational: a held stock without a bar; any nonzero reconciliation difference; an actual fill differing from the modeled fill. Performance: drawdown above 8% of the equity peak; observed costs differing from modeled costs by more than 50% over 20 fills. A trigger starts a review; any rule change becomes a new version.	Teaching setting
Overdue time exit after halt	First resumed open: opening stop/target gap first; otherwise overdue time exit at the open, before intraday tests. Keep stale marks and occupied slot until then.	Convention

Replay inputs: completed S4 and S5 bars (O/H/L/C, fictional USD)

Session	Fir	Alder	Birch	Dogwood	Cedar
S4	33.10/33.60/32.40/33.20	33.80/34.70/33.50/34.40	73.00/73.50/72.20/72.90	62.30/63.80/62.00/63.50	43.20/43.90/42.80/43.60
S5	Not supplied; first determine whether S4 leaves it held	34.50/35.60/34.30/35.40	72.90/76.60/70.00/71.20	63.40/64.10/62.60/62.90	43.50/44.20/43.10/43.30

Replay inputs: screen facts released at each decision close

Decision close	Unheld candidate / close ($)	Prior-20 high ($)	20-session mean dollar volume	Next earnings session	History completeness
D4	No non-held stock signals	—	—	—	Supplied screen covers every non-held stock
D5	Ginkgo / 25.40	25.00	Not computed	44	S3 bar missing in required history

Compare the continuation with your frozen note. This reference includes outcome headings, complete ledgers and solutions. In the learner print edition it starts on a new page after the cutoff packet.

Start with an economic question

A trading system is an organized set of decisions that can be applied consistently. It need not be automated, but its rules should be clear enough to evaluate. Begin with a proposed mechanism and an observable claim. “Trends may persist over this horizon” is a research hypothesis. “This indicator predicts everything” is not a useful specification. The mechanism does not prove an edge; it tells you what evidence would support or weaken the idea.

Earlier chapters described a fictional research system called Atlas-20, which studies daily stock breakouts, as a list of decisions a designer must make. This chapter makes every one of those decisions. It freezes version 1.0 of the rule sheet, supplies a small invented market of eight stocks over eight sessions, and processes each decision in order until the ledger reconciles to the cent. All companies, prices, volumes, and calendars are fictional. The percentages are teaching settings, not recommended allocations, and the system has no demonstrated performance. The scenario was built to exercise the rules, with a tie, a partial fill, an unfilled order, a halt, a gap, an ambiguous bar, and a time exit, so it is not a typical week.

Use one test for completeness: two careful readers given the same inputs should produce the same ledger without asking the author anything. Each rule below is labeled as a definition, a modeling convention, a scope choice, or an arbitrary teaching setting. The included scripts/atlas20-sim.cjs processes the supplied scenario under the rule sheet and reproduces its accounting; it is a teaching simulator, not a trading engine. You can reproduce them by hand.

Write the data and eligibility contract

Atlas-20 uses a point-in-time universe and completed regular-session daily bars. “Point in time” means the system uses the member list, prices, and calendar as they were known at each decision close, not as they are known today. The signal needs the twenty sessions before the decision day, so a stock needs 21 consecutive valid bars before it can be screened. If any of those bars is missing, v1.0 records an incomplete record and evaluates nothing that day. It does not fill the gap with yesterday’s price, because an invented bar could create or hide a breakout.

Each filter is part of the research claim, so each is stated with its data and timing. The liquidity filter averages close × volume over the twenty sessions ending at the decision close. A cutoff chosen from the full sample would quietly use the future. The price band keeps the fixed $2

stop between 2% and 10% of the share price. Without it, the same rule would mean very different risks for a $15 stock and a $150 stock. The earnings exclusion covers sessions t+1 through t+10 because this is the planned ten-session holding window for an entry in t+1 (holding session 1), ending at t+10 (holding session 10), so entries exclude announcements known at the decision close within that window. A rescheduled or unexpected announcement remains possible; the calendar is not insurance. The checks run in a declared order, and the first failure is recorded. That keeps the record of skipped candidates honest. A halt can extend the actual elapsed holding beyond that planned window; the calendar filter cannot exclude every announcement during an overdue holding.

Atlas-20 v1.0 rule sheet, part 1: data, universe, and eligibility

Rule	Atlas-20 v1.0 setting	Kind
Scope	Long-only fictional U.S.-listed common stocks in one U.S.-dollar account; no leverage, short sales, or derivatives.	Scope choice
Universe	Point-in-time member list as of each decision date. Scenario members throughout: Alder, Birch, Cedar, Dogwood, Elm, Fir, Ginkgo, Hazel.	Convention
Data and clock	Completed regular-session daily OHLC and share volume; split adjustment uses only actions effective by the decision date. Decisions are computed after the close; orders are submitted before the next open.	Convention
Minimum history	21 consecutive valid bars: sessions t−20 through t.	Teaching setting
Missing data	Any absent or invalid bar in t−20…t: no signal is evaluated that day. No interpolation or carried-forward bars.	Convention
Already held	A stock with an open position or a working order is not screened; v1.0 never adds to a position.	Teaching setting
Signal	Close at t strictly above the highest high of sessions t−20…t−1.	Definition
Price band	$20.00 ≤ decision close ≤ $100.00, so the fixed $2 stop is 2%–10% of price.	Teaching setting

Rule	Atlas-20 v1.0 setting	Kind
Liquidity	Mean of close × volume over sessions t−19…t ≥ $2,000,000 (past data only).	Teaching setting
Earnings	No scheduled earnings in sessions t+1…t+10 (the planned ten-session holding window; a halt can delay exit), per the calendar snapshot at the decision close.	Teaching setting
Regime filter	None in v1.0.	Scope choice
Check order	Complete record → signal → price band → liquidity → earnings. The first failure is recorded.	Convention

Translate a signal into portfolio decisions

An eligible stock signals when its completed close exceeds the highest high of the prior twenty completed sessions. When several qualify at the same close, Atlas-20 ranks them by breakout margin, the percentage by which the close clears that high. Rank the margin rounded to 0.01 percentage point, as stated in the table; ties in that rounded value are broken alphabetically. Two unequal raw margins can therefore receive the same rank value. Do not silently switch to unrounded ranking. This rule is not wise or optimized. Its only virtue is that it is written down. Without a declared order, a backtest quietly selects candidates by file order or by whichever result the researcher noticed first.

At most five slots exist, and a slot is occupied by any open position, including a halted one, and by any order placed at the current close. Each order's quantity is the smallest of three allowances computed from equity E at the decision close. The risk allowance spends at most 0.5% of E on planned price risk at $2 per share. The notional allowance keeps the position at or below 10% of E at the limit price. The cash allowance uses only cash not already reserved for an order placed earlier at the same close. Following the book's contract, 1R is the frozen planned price risk: filled shares × $2.00, fixed at the fill. The $5 round-trip cost is shown separately. So an ordinary stop exit for a full 250-share position loses $500 of price and $505 including costs. That cost-inclusive figure is a separate, labeled quantity, not 1R.

The cash test deserves a remark. Five slots at a 10% notional cap can commit only about half of equity, so cash does not bind in this unleveraged, mildly varying scenario. Large mark changes, stale valuations, or a later version can make the cash test important; the entry caps do not bound subsequent market values. It is still computed at every decision, for two reasons. Reservations must prevent the same dollars from funding two orders, and any later version

with larger caps or more slots can make cash bind. One of the practice tasks shows such a variant.

Atlas-20 v1.0 rule sheet, part 2: ranking, slots, sizing, and costs

Rule	Atlas-20 v1.0 setting	Kind
Ranking value	Breakout margin = 100 × (close − prior-20 high) ÷ prior-20 high, rounded to 0.01 percentage point; larger ranks first.	Teaching setting
Tie-break	Alphabetical by company name.	Convention
Slots	At most 5 = open positions (halted ones included) + orders placed at this close.	Teaching setting
Sizing equity E	Cash + shares × close for every holding, after the session’s exits and fills; a halted holding uses its last valid close.	Definition
Limit price L	1.01 × signal close, rounded down to the cent.	Teaching setting
Risk allowance	floor(0.5% × E ÷ $2.00) shares.	Teaching setting
Notional allowance	floor(10% × E ÷ L) shares.	Teaching setting
Cash allowance	floor((cash − cash already reserved at this close − $2.50) ÷ L) shares.	Definition
Quantity	Smallest of the three allowances; fewer than 1 share means no order.	Definition
Cash reservation	Shares × L + $2.50, held from order placement until the order fills or expires.	Convention
R	1R = filled shares × $2.00, frozen at the fill. Net R = net P&L ÷ 1R. Costs are shown separately.	Book contract
Costs	$2.50 per executed fill (entry, exit, or partial entry), so $5.00 per round trip. No separate spread or slippage charge beyond the fill rules.	Teaching setting
Cash interest and settlement	Cash earns zero; settlement timing is ignored.	Convention

Exits, gaps, and time are first-class rules

Entries use a day-limit buy order for the next session at 1.01 × the signal close, rounded down to the cent. Investor.gov explains that a buy limit executes only at its limit or lower and is not guaranteed to execute, and that a day order generally cancels at the end of the regular session. Atlas-20 models those facts with deliberately conservative conventions. An open at or below the limit fills at the open. A later trade strictly below the limit fills at the limit. A low exactly equal to the limit shows only that the price was touched, which does not prove that shares were available, so it is not a fill. The filled quantity is capped at 1% of the session's share volume, and the rest of the order is canceled.

Each fill receives a sell stop $2.00 below and a sell limit target $4.00 above the actual fill price, plus a time exit at the close of the tenth holding session. Investor.gov notes that a stop becomes a market order once its price is reached and can execute well away from that price. Atlas-20 therefore models an ordinary stop at the stop price, an opening gap through the stop at the open, and a bar that contains both levels as a stop. Daily data cannot show which level traded first, and assuming the favorable order would flatter the record. These are assumptions about fills, not reports of actual executions.

A trading halt is a different problem. According to Investor.gov, the most common regulatory halt occurs when news is pending, and the security's primary exchange decides whether to impose it. While a held stock has no bar, no exit can be modeled. Atlas-20 marks the stock at its last valid close, flags that mark as stale, keeps the slot occupied, and applies the gap rule to the first new bar. Cash earns nothing, settlement timing is ignored, and short sales, leverage, and derivatives are outside the scope of v1.0.

The 1% volume cap is a retrospective daily-bar fill model: the total session volume is unknown at the opening trade. It cannot be used as an instruction for how many shares to execute at that open. A live participation algorithm would need intraday volume and quote data, which this scenario does not provide. There are no splits, dividends, membership changes or calendar revisions during these eight sessions; a deployment specification would need separate policies for them. The supplied rolling-high, liquidity and completeness fields are sufficient statistics for this exercise, not a full historical feed.

If a halt spans holding-session-10 close, record an overdue time exit and retain the slot and stale mark. At the first subsequent tradable opening, opening stop/target gap rules have priority; otherwise exit at that open for the overdue deadline, before considering that bar's intraday range. For entry E1 and a halt at E10–E12, a reopening at E13 executes this overdue rule. E10 remains the scheduled date; E13 is the later modeled execution date.

Atlas-20 v1.0 rule sheet, part 3: orders, fills, exits, and review

Rule	Atlas-20 v1.0 setting	Kind
Entry order	Buy day-limit order at L for the next session only; any unfilled remainder cancels at that session's close.	Convention
Entry fill	Open ≤ L: fill at the open. Otherwise, low strictly below L: fill at L. A low exactly at L is a touch, not a fill.	Convention (conservative)
Partial fill	Filled shares ≤ floor(1% × entry-session volume). Keep the filled shares, cancel the rest, never top up.	Teaching setting
Rejected or unfilled candidates	Not re-offered. A stock needs a new signal at a later close.	Convention
Stop	Sell stop at fill − $2.00; triggered when low ≤ stop.	Teaching setting
Target	Sell limit at fill + $4.00; fills only when high is strictly above the target.	Teaching setting
Time exit	At the close of holding session 10 (entry session = 1; exchange sessions counted, including halted ones).	Teaching setting
Exit priority (held from an earlier session)	1. Open ≤ stop: exit at open. 2. Open ≥ target: exit at open. 3. Low ≤ stop and high > target: stop assumed first. 4. Low ≤ stop: exit at stop. 5. High > target: exit at target. 6. Holding session 10: exit at close. After opening checks 1–2, an overdue time deadline exits at the open before steps 3–5; a scheduled session-10 close uses step 6.	Convention (conservative)
Entry bar	Fill at the open: steps 3–5 apply to the rest of that bar. Fill at L intraday: the stop applies if low ≤ stop; a target is not credited.	Convention (conservative)
Halt or missing bar while held	No exit is possible. Mark at the last valid close, flagged stale; the position keeps its slot. The first new bar starts at step 1.	Convention

Rule	Atlas-20 v1.0 setting	Kind
Processing order	Batch accounting: held-position exits and reserved entries, then close marks and screening. Entry quantities are fixed at the previous close and fully cash-reserved; no same-session exit funds a larger order. Within-session event rows are not a timestamped tape.	Convention
Review triggers	Operational: a held stock without a bar; any nonzero reconciliation difference; an actual fill differing from the modeled fill. Performance: drawdown above 8% of the equity peak; observed costs differing from modeled costs by more than 50% over 20 fills. A trigger starts a review; any rule change becomes a new version.	Teaching setting
Overdue time exit after halt	First resumed open: opening stop/ target gap first; otherwise overdue time exit at the open, before intraday tests. Keep stale marks and occupied slot until then.	Convention

FICTIONAL TEACHING ILLUSTRATION

Atlas-20 daily processing order

1. Carry in the frozen state

Rule sheet v1.0, holdings with their stops, targets and holding-session counts, orders placed at the previous close, and reserved cash.

↓

2. Process held positions

No bar: mark stale and keep the slot. Otherwise apply the exit priority: gap at the open, both levels in one bar (stop first), stop, target, then the time exit at the close.

↓

3. Process entry orders

Open at or below the limit fills at the open; a trade below the limit fills at the limit; a touch or a higher low cancels. Cap shares at 1% of volume and release every reservation.

↓

4. Mark the close

Equity = cash + shares × close (halted: last valid close). Check that cash flows reconcile to the cent.

↓

5. Screen

Skip held stocks. Then check the complete 21-bar record, the signal, the price band, liquidity, and the earnings window, recording the first failure.

↓

6. Rank, size, reserve

Sort by breakout margin, then name. Free slots = 5 − open positions. Shares = the smallest of the risk, notional, and cash allowances; reserve shares × limit + $2.50.

↓

7. Log and review

Record orders, rejections, skips, and any review trigger before the next open. A trigger starts a review; it does not change the rules.

Fixed daily-bar accounting sequence for the fictional scenario. Orders and cash reservations are frozen at the prior close. Batch processing of a close exit before an opening entry is an accounting order only; no later proceeds fund that entry or increase its quantity.

View the illustration’s data

The supplied scenario

The account begins at the session-0 close with $100,000.00 of equity: $81,330.00 in cash and two positions entered before the scenario. Hazel's tenth holding session is session 2; Fir's would be session 7. The screen log is supplied data. It lists every stock that was not held and whose close exceeded its prior-20 high at a decision close, along with the pre-computed liquidity, the calendar, and the completeness facts. Every unlisted non-held stock and decision-close pair did not signal. Held stocks are excluded before screening; their omission says nothing about whether their close exceeded a rolling high. The two price grids supply each bar the rules need. A dash means the bar is not needed because the stock was neither held nor being bought. "No bar" means the record does not exist.

Before reading on, apply the rule sheet to decision close 0 yourself. List the six signals, decide which are eligible, rank them, and size whatever fits in the free slots.

Opening state at the session-0 close

Holding	Shares	Entry session	Fill ($)	Stop / target ($)	1R ($)	Session-0 close ($)	Value ($)
Hazel	200	−7 (holding session 10 = session 2)	47.00	45.00 / 51.00	400.00	48.60	9,720.00
Fir	250	−2 (holding session 10 = session 7)	36.00	34.00 / 40.00	500.00	35.80	8,950.00
Cash	—	—	—	—	—	—	81,330.00
Equity	—	—	—	—	—	—	100,000.00

Screen log: every non-held stock whose close exceeded its prior-20 high

Decision close	Stock	Close ($)	Prior-20 high ($)	20-session mean dollar volume ($ million)	Next scheduled earnings (session)	21-bar record
D0	Alder	31.96	31.00	9.8	31	complete
D0	Birch	71.75	70.00	2.3	40	complete
D0	Cedar	40.80	40.00	6.4	25	complete
D0	Dogwood	61.20	60.00	14.1	35	complete
D0	Elm	55.40	54.00	11.7	6	complete

Decision close	Stock	Close ($)	Prior-20 high ($)	20-session mean dollar volume ($ million)	Next scheduled earnings (session)	21-bar record
D0	Ginkgo	24.60	24.10	1.6	44	complete
D1	Dogwood	62.40	61.50	14.3	35	complete
D2	Cedar	42.30	41.95	6.6	25	complete
D3	Hazel	51.80	50.90	5.2	38	complete
D5	Ginkgo	25.40	25.00	not computed	44	session-3 bar missing
D6	Elm	58.90	56.30	13.9	68	complete

Supplied bars, part 1 (O/H/L/C in $; volume in shares where an entry needs it)

Session	Alder	Birch	Cedar	Dogwood
0	31.10/32.05/31.00/31.96	70.40/71.90/70.10/71.75	40.10/41.60/40.00/40.80	60.30/61.50/60.10/61.20
1	32.10/32.60/31.70/32.45; vol 410,000	72.20/73.10/71.60/72.80; vol 11,800	41.60/41.95/41.20/41.50; vol 240,000	61.40/62.60/61.20/62.40
2	32.50/33.40/32.20/33.20	72.90/73.60/72.10/73.30	41.70/42.40/41.40/42.30	62.80/63.30/62.10/62.90; vol 520,000
3	33.30/34.10/33.00/33.90	73.20/74.00/72.60/73.10	43.00/43.40/42.50/43.10; vol 260,000	62.70/63.20/61.90/62.20
4	33.80/34.70/33.50/34.40	73.00/73.50/72.20/72.90	43.20/43.90/42.80/43.60	62.30/63.80/62.00/63.50
5	34.50/35.60/34.30/35.40	72.90/76.60/70.00/71.20	43.50/44.20/43.10/43.30	63.40/64.10/62.60/62.90
6	35.50/36.40/35.20/36.00	—	43.40/44.00/42.70/43.80	62.80/63.40/61.70/62.10
7	—	—	43.90/44.60/43.50/44.30	62.20/63.60/61.90/63.30

Supplied bars, part 2 (— = not needed; no bar = record absent)

Session	Elm	Fir	Ginkgo	Hazel
0	54.20/55.60/54.00/55.40	35.90/36.20/35.40/35.80	24.00/24.70/23.90/24.60	48.20/48.90/47.90/48.60
1	—	35.70/35.95/34.60/34.90	—	48.70/49.40/48.30/49.10

Session	Elm	Fir	Ginkgo	Hazel
2	—	34.80/35.30/34.40/35.10	—	49.20/49.80/48.90/49.30
3	—	no bar	no bar	49.60/51.95/49.50/51.80
4	—	33.10/33.60/32.40/33.20	—	—
5	—	—	25.00/25.50/24.90/25.40	—
6	57.00/59.20/56.80/58.90	—	—	—
7	59.10/59.90/58.70/59.60; vol 300,000	—	—	—

Decision close 0: four eligible candidates, three slots

Six stocks signal. Elm is excluded because its scheduled earnings fall in session 6, inside the window from session 1 to session 10. Ginkgo is excluded because its mean dollar volume of $1.6 million is below the $2.0 million threshold. Both would have looked attractive on a chart, and both stay in the opportunity record as ineligible. The remaining four rank as Alder at 3.10%, Birch at 2.50%, and then Cedar and Dogwood tied at exactly 2.00%. Cedar wins the tie alphabetically. Hazel and Fir occupy two of the five slots, so only three are free, and Dogwood is recorded as “eligible, no free slot.”

At E = $100,000.00, the risk allowance is floor(500 ÷ 2) = 250 shares for every candidate. Alder’s limit is $32.27, so its notional allowance is floor(10,000 ÷ 32.27) = 309, and risk binds at 250 shares. Birch’s limit is $72.46, so notional binds at 138 shares, which is $9,999.48 at the limit. Cedar’s notional allowance is 242 shares at $41.20. The cash allowances, computed in rank order after earlier reservations, are far larger. The three reservations total $28,044.88 and leave $53,285.12 available. The same price-risk rule produces a full-risk Alder order and two smaller orders because the notional cap binds for the higher-priced stocks.

Decision close 0: screening, ranking, and sizing (E = $100,000.00; 3 free slots)

Candidate	Margin (%)	Eligibility	Rank	Limit L ($)	Risk / notional / cash allowance (shares)	Order (shares)	Reserved ($)
Alder	3.10	Yes	1	32.27	250 / 309 / 2,520	250 (risk binds)	8,070.00

Candidate	Margin (%)	Eligibility	Rank	Limit L ($)	Risk / notional / cash allowance (shares)	Order (shares)	Reserved ($)
Birch	2.50	Yes	2	72.46	250 / 138 / 1,011	138 (notional binds)	10,001.98
Cedar	2.00	Yes	3	41.20	250 / 242 / 1,535	242 (notional binds)	9,972.90
Dogwood	2.00	Yes	4	61.81	250 / 161 / 862	0: no free slot	—
Elm	2.59	No: earnings in session 6	—	—	—	—	—
Ginkgo	2.07	No: $1.6M < $2.0M	—	—	—	—	—
Total	—	—	—	—	—	3 orders	28,044.88

Sessions 1–3: fills, a nonfill, a time exit, and a halt

Session 1 begins with the held positions. Hazel's range stays between its $45 stop and $51 target. Fir's low of $34.60 stays above its $34.00 stop. Next come the entries. Alder opens at $32.10, below its $32.27 limit, so all 250 shares fill at the open. Its stop is $30.10, its target is $36.10, and 1R is $500. Birch opens at $72.20, within its limit, but only 11,800 shares trade, so the 1% cap allows 118 of the 138 shares ordered. The remainder cancels, and Birch's 1R is 118 × $2 = $236, not the $276 planned at the order. Cedar opens at $41.60, above its $41.20 limit, and its low is exactly $41.20. That is a touch, so there is no fill, and the $9,972.90 reservation is released. Cash falls from $81,330.00 by $8,027.50 and $8,522.10 to $64,780.40, and equity at the close is $100,028.30.

Cedar's unused slot does not go back to Dogwood, because candidates are offered only at their own signal close. Dogwood earns a new chance by signaling again at decision close 1: its $62.40 close exceeds the new prior-20 high of $61.50. The 158-share order reflects the notional allowance, floor(10,002.83 ÷ 63.02). In session 2, Dogwood opens at $62.80, below its limit, and fills there. Hazel reaches holding session 10 without touching either level and exits at the $49.30 close for +$455 net, or +1.14R on its $400 denominator. At decision close 2, Cedar signals again at $42.30, above the $41.95 high it set in session 1, and receives a 234-share order at $42.72.

In session 3, Fir has no bar: trading is halted. Fir is marked at its session-2 close of $35.10 and flagged stale, and the missing bar fires an operational review trigger. Cedar opens at $43.00, above its limit, then trades down to $42.50, so the model fills it at $42.72. Because the fill happened intraday, the entry bar cannot credit a target, and the stop is not reached. At the close, five slots are in use: Fir, Alder, Birch, Dogwood, and Cedar. Hazel then signals at $51.80 and passes every filter, but the halted Fir still holds a slot, so Hazel gets no order. Equity is $100,502.82, and holdings are 45.6% of it, the highest exposure in the scenario.

Pause here. The main exercise asks you to process sessions 4 and 5 from this state before you look at the complete ledger in the next section.

Sessions 4–7 and the complete ledger

Fir resumes trading in session 4 with an open of $33.10, below its $34.00 stop, so the exit occurs at the open. The loss is −$725 of price and −$730 net, or −1.46R. That is well beyond the ordinary $505 cost-inclusive modeled loss, because the gap skipped the stop price. In session 5, Birch trades from $70.00 to $76.60, a bar that contains both its $70.20 stop and its $76.20 target. The conservative priority records a stop: −$241 net, or −1.02R. A reader who assumed the target came first would report +$467, or +1.98R. A single unverifiable assumption about the order of trades moves this result by three R. At decision close 5, Ginkgo's close is above its prior-20 high, but its session-3 bar is missing from the record, so nothing is evaluated.

Alder trades above its $36.10 target in session 6 and exits there for +$995 net, or +1.99R. At that close, Elm signals again. Its earnings date of session 6 is now in the past, and its next scheduled report falls outside the window. It receives a 168-share order, which fills at the $59.10 open of session 7. The scenario ends with Dogwood, Cedar, and Elm open and marked at the session-7 close.

Every row of the ledger satisfies cash + holdings = equity. Equity rises from $100,000.00 to $100,739.22, a change of $739.22. Measured from the session-0 marks, the closed trades contribute +$137.50 for Hazel, −$677.50 for Fir, −$241.00 for Birch, and +$995.00 for Alder, a total of $214.00. The three open positions contribute $532.72 of unrealized price gain less $7.50 of entry costs, or $525.22. The total, $214.00 + $525.22, equals the $739.22 change. Nine fills cost $22.50 inside the scenario.

Session event log (cash flows include the $2.50 fill cost)

Session	Event	Cash flow ($)
0	Opening state; orders placed at the close (see allocation table)	—

Session	Event	Cash flow ($)
0	Close marks: Hazel 200 × 48.60 = 9,720.00; Fir 250 × 35.80 = 8,950.00	—
1	Alder buy 250 @ $32.10 (open); stop $30.10, target $36.10	−8,027.50
1	Birch buy 118 of 138 (partial: cap 118) @ $72.20 (open); stop $70.20, target $76.20	−8,522.10
1	Cedar buy 242 @ limit $41.20: no fill; $9,972.90 released	0.00
1	Close marks: Hazel 200 × 49.10 = 9,820.00; Fir 250 × 34.90 = 8,725.00; Alder 250 × 32.45 = 8,112.50; Birch 118 × 72.80 = 8,590.40	—
2	Hazel exit 200 @ $49.30 (time exit at close (holding session 10))	9,857.50
2	Dogwood buy 158 @ $62.80 (open); stop $60.80, target $66.80	−9,924.90
2	Close marks: Fir 250 × 35.10 = 8,775.00; Alder 250 × 33.20 = 8,300.00; Birch 118 × 73.30 = 8,649.40; Dogwood 158 × 62.90 = 9,938.20	—
3	Fir halted: no bar; marked at last close $35.10	0.00
3	Cedar buy 234 @ $42.72 (intraday limit fill); stop $40.72, target $46.72	−9,998.98
3	Close marks: Fir 250 × 35.10 (stale) = 8,775.00; Alder 250 × 33.90 = 8,475.00; Birch 118 × 73.10 = 8,625.80; Dogwood 158 × 62.20 = 9,827.60; Cedar 234 × 43.10 = 10,085.40	—
4	Fir exit 250 @ $33.10 (gap through stop, exit at open)	8,272.50
4	Close marks: Alder 250 × 34.40 = 8,600.00; Birch 118 × 72.90 = 8,602.20; Dogwood 158 × 63.50 = 10,033.00; Cedar 234 × 43.60 = 10,202.40	—

Session	Event	Cash flow ($)
5	Birch exit 118 @ $70.20 (stop and target in one bar: stop assumed first)	8,281.10
5	Close marks: Alder 250 × 35.40 = 8,850.00; Dogwood 158 × 62.90 = 9,938.20; Cedar 234 × 43.30 = 10,132.20	—
6	Alder exit 250 @ $36.10 (target)	9,022.50
6	Close marks: Dogwood 158 × 62.10 = 9,811.80; Cedar 234 × 43.80 = 10,249.20	—
7	Elm buy 168 @ $59.10 (open); stop $57.10, target $63.10	−9,931.30
7	Close marks: Dogwood 158 × 63.30 = 10,001.40; Cedar 234 × 44.30 = 10,366.20; Elm 168 × 59.60 = 10,012.80	—

Close-of-session accounting (cash + holdings = equity)

Session	Cash ($)	Holdings at close ($)	Equity ($)	Open positions	Orders placed at this close	Reserved ($)	Available cash ($)
0	81,330.00	18,670.00	100,000.00	2	3 orders; Dogwood: no slot; Elm, Ginkgo ineligible	28,044.88	53,285.12
1	64,780.40	35,247.90	100,028.30	4	Dogwood 158 @ 63.02	9,959.66	54,820.74
2	64,713.00	35,662.60	100,375.60	4	Cedar 234 @ 42.72	9,998.98	54,714.02
3	54,714.02	45,788.80	100,502.82	5 (Fir stale)	None: Hazel eligible, no free slot	0.00	54,714.02
4	62,986.52	37,437.60	100,424.12	4	None: no signal	0.00	62,986.52
5	71,267.62	28,920.40	100,188.02	3	None: Ginkgo record incomplete	0.00	71,267.62

Session	Cash ($)	Holdings at close ($)	Equity ($)	Open positions	Orders placed at this close	Reserved ($)	Available cash ($)
6	80,290.12	20,061.00	100,351.12	2	Elm 168 @ 59.48	9,995.14	70,294.98
7	70,358.82	30,380.40	100,739.22	3	None: no signal (scenario ends)	0.00	70,358.82

Closed trades (net = gross − $5.00 round-trip cost; R on the frozen 1R)

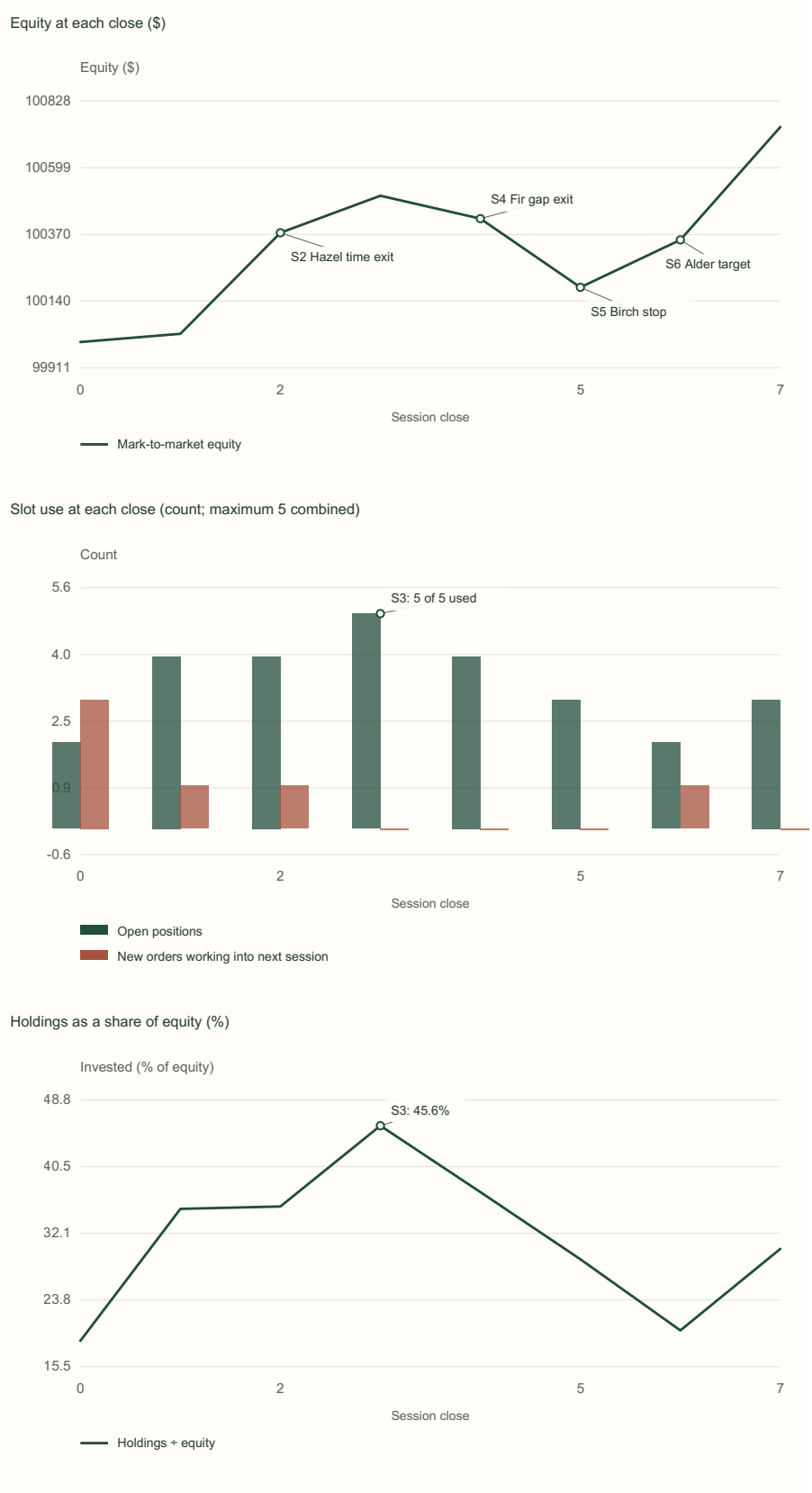
Trade	Shares	Entry → exit	Exit reason	1R ($)	Gross / costs / net ($)	Net R
Hazel	200	S-7 @ 47.00 → S2 @ 49.30	time exit at close	400.00	460.00 / 5.00 / 455.00	+1.14
Fir	250	S-2 @ 36.00 → S4 @ 33.10	gap through stop: open	500.00	−725.00 / 5.00 / −730.00	−1.46
Birch	118	S1 @ 72.20 → S5 @ 70.20	both levels in bar: stop first	236.00	−236.00 / 5.00 / −241.00	−1.02
Alder	250	S1 @ 32.10 → S6 @ 36.10	target	500.00	1,000.00 / 5.00 / 995.00	+1.99

Open positions at the session-7 close

Position	Shares	Fill ($)	Stop / target ($)	1R ($)	Session-7 close ($)	Unrealized gross ($)
Dogwood	158	S2 @ 62.80	60.80 / 66.80	316.00	63.30	79.00
Cedar	234	S3 @ 42.72	40.72 / 46.72	468.00	44.30	369.72
Elm	168	S7 @ 59.10	57.10 / 63.10	336.00	59.60	84.00

FICTIONAL TEACHING ILLUSTRATION

Atlas-20 scenario: equity, slots, and exposure at each close



Fictional Atlas-20 v1.0 scenario, read from the close-of-session ledger; session 0 is the opening state. Equity moves by less than 1% while exposure never exceeds 45.6%,

under the supplied path and entry caps. Subsequent prices can change exposure; the cap is applied at order placement. Look at session 3, where all five slots are used even though Fir is halted.

View the illustration’s data

Equity at each close ($)

Session close	Mark-to-market equity
0	100000
1	100028.3
2	100375.6
3	100502.82
4	100424.12
5	100188.02
6	100351.12
7	100739.22

Slot use at each close (count; maximum 5 combined)

Session close	Open positions	New orders working into next session
0	2	3
1	4	1
2	4	1
3	5	0
4	4	0
5	3	0
6	2	1
7	3	0

Holdings as a share of equity (%)

Session close	Holdings ÷ equity
0	18.7
1	35.2

Session close	Holdings ÷ equity
2	35.5
3	45.6
4	37.3
5	28.9
6	20
7	30.2

What the ledger can and cannot tell you

The +0.74% change over seven sessions is evidence of nothing about Atlas-20's merit. It comes from four closed trades and three open ones, invented to exercise the rules. What the ledger does show is structural. Exposure peaked at 45.6% in this supplied path. Entry caps constrain orders at their limits; subsequent mark changes can move holdings above those percentages. Slots, not cash, rationed the opportunities: Dogwood at decision close 0 and Hazel at decision close 3 were eligible and turned away. The conventions also matter. The no-fill on a touch, the participation cap, and the stop-first rule each made the record less favorable than a looser reading would have.

The ledger also keeps the decisions that never became trades. Elm and Ginkgo were ineligible, Cedar did not fill, Dogwood and Hazel found no free slot, and Ginkgo's record was incomplete. Testing, discussed in the earlier evidence chapter, needs exactly this full opportunity set. A researcher who keeps only the trades cannot ask whether the ranking rule, the slot limit, or the earnings window is doing useful work. Answering those questions takes many more opportunities than this scenario provides, held-out data, and realistic costs. The case studies in the last chapter return to such records.

A regime filter would be version 1.1

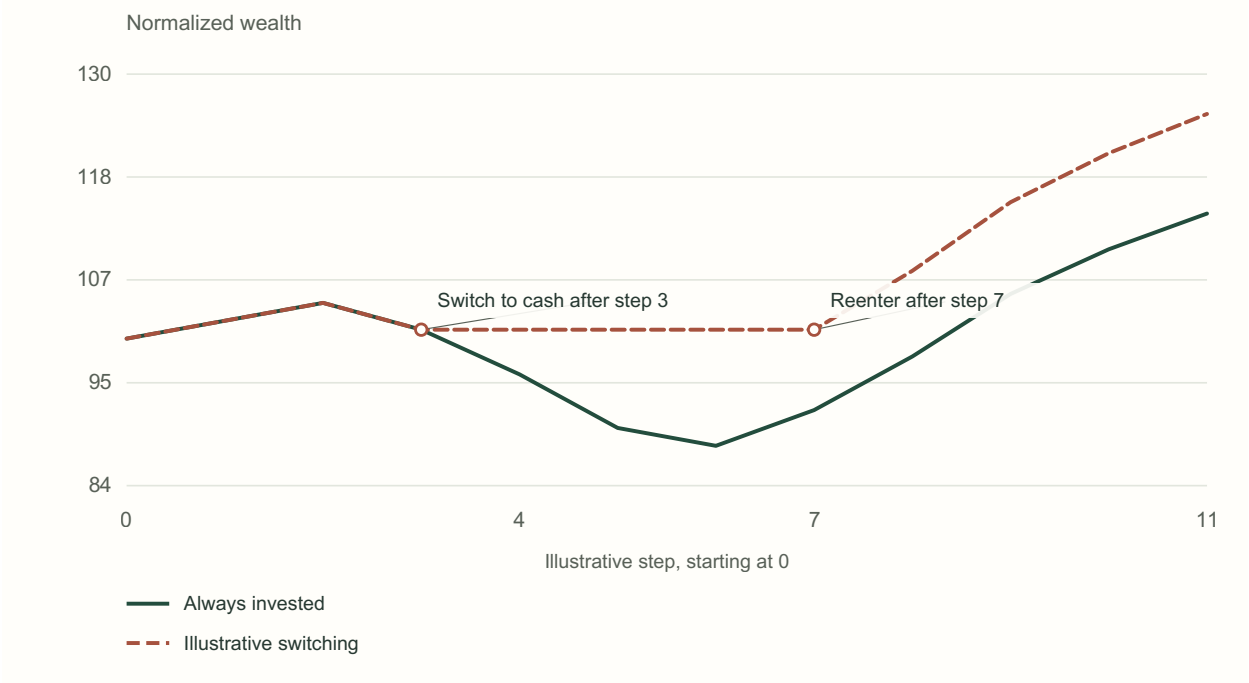
A regime filter restricts trading according to a defined market state. Atlas-20 v1.0 has none. Adding one creates a new version that needs its own data contract. A complete specification might read: "No new entries at a decision close unless the fictional benchmark's completed close exceeds the simple average of its previous 50 completed closes. Existing positions are unaffected. If the benchmark bar is missing, no new entries." That sentence names the series, the lookback, the timestamp, what the filter blocks, and its missing-data rule. It is still a

hypothesis, not an improvement. A filter can skip losing periods and also miss rebounds. If its definition is repeatedly altered to fit the same history, the search grows while the final system appears more careful.

The separate fictional figure contrasts two predetermined exposure schedules over an invented price path. The switching path holds 101 in cash for steps 4–7 and reenters at an index of 92, so its later values equal 101 × index ÷ 92. The figure shows the consequences of exposure timing, not evidence that any filter can identify those dates. In a real test, every switch must follow information available at the time, and cash returns and costs must be included. Measuring only the invested periods would hide the opportunity cost of sitting in cash.

FICTIONAL TEACHING ILLUSTRATION

Exposure timing changes a path (separate illustration)



Separate fictional illustration, not part of the Atlas-20 scenario. Normalized wealth starts at step 0. The switching path follows the stock through step 3, holds 101 in cash for steps 4–7, and reenters after step 7 at stock index 92; later values equal 101 × (index ÷ 92), rounded to two decimals. Cash earns zero and costs are omitted. The switch dates are given, not produced by a tested filter.

View the illustration’s data

Exposure timing changes a path (separate illustration)

Illustrative step, starting at 0	Always invested	Illustrative switching
0	100	100
1	102	102
2	104	104
3	101	101
4	96	101
5	90	101
6	88	101
7	92	101
8	98	107.59
9	105	115.27
10	110	120.76
11	114	125.15

Specify when to pause and how to improve

A monitoring plan compares observed behavior with the specification: signal counts, data completeness, exposure, costs, and differences between modeled and actual fills. Atlas-20 separates operational triggers from performance triggers. Fir's missing bar in session 3 fired an operational trigger. The right response is to confirm the halt and the data feed, not to change a parameter. The scenario's largest decline from an equity peak, from $100,502.82 to $100,188.02, is about 0.31%, far from the 8% performance trigger. Even a breached trigger only starts a review; a losing week is not proof that the hypothesis has failed.

Any change to the rule sheet, whether a new ranking, a different cap, or a regime filter, becomes a new version with its own record. Bailey and coauthors explain why this matters. The chance of a false discovery rises with the number of configurations tried on the same data, and published backtests rarely report how many trials were run. Keep every version and the reason for it, evaluate on data not used to choose the change, and compare with simpler alternatives. A complete system is valuable because it exposes every decision to scrutiny. It becomes credible only through honest evidence, feasible execution, and a risk policy suited to its uncertainty.

Atlas-20 still needs its own empirical test

The research appendix gives supportive momentum evidence and adverse rule-selection evidence under other specifications. Atlas-20's thresholds, allocation and fills are original teaching choices. The cash reconciliation proves that the scenario's accounting works; it is not a held-out real-stock trial or an estimate of its profitability.

A WORKED EXAMPLE

Two limits disagree

At decision close 0, equity is $100,000.00 and the risk allowance is 250 shares for every candidate. Alder's $32.27 limit allows 309 shares under the 10% notional cap, so risk binds at 250 shares: $8,067.50 of notional and a 1R of $500. Birch's $72.46 limit allows only 138 shares, so notional binds. Its planned risk is $276, but after the partial fill of 118 shares its frozen 1R is $236. The qualifying signal does not override the tighter limit, and the denominator follows the shares actually filled.

Pause & try it

Start from the session-3 close. Cash is $54,714.02; the holdings are Fir 250 shares (halted, last close $35.10, stop $34.00), Alder 250 (stop $30.10, target $36.10), Birch 118 (stop $70.20, target $76.20), Dogwood 158 (stop $60.80, target $66.80), and Cedar 234 (stop $40.72, target $46.72); no orders are working. Using the rule sheet, the price grids, and the screen log, process session 4, decision close 4, session 5, and decision close 5. For each, record every exit price and reason, cash after the fills, each close mark, equity, free slots, and any order. Then give the net P&L and net R of the Fir and Birch trades.

GUIDED PRACTICE

Rerank decision close 0 by liquidity

Keep every v1.0 rule except the ranking: rank eligible candidates by 20-session mean dollar volume, largest first. Using the screen log, which three candidates receive orders

at decision close 0, how many shares each, and how much cash is reserved? Hint: rank first, then size in rank order, deducting each reservation before computing the next cash allowance.

INDEPENDENT PRACTICE

A variant in which cash binds

A deliberately aggressive teaching variant, for arithmetic only, raises the risk budget to 2% of equity and the notional cap to 40%, with every other v1.0 rule unchanged. Redo the decision-close-0 sizing for Alder, Birch, and Cedar. Which allowance binds for each, and how much cash is left after the three reservations?

INDEPENDENT PRACTICE

Diagnose a substituted order

A colleague’s ledger shows Dogwood bought during session 1 at its decision-close-0 limit of $61.81. The colleague reasons that Cedar failed to fill, so its slot should pass to the next-ranked candidate. The session-1 open of $61.40 then fills Dogwood below the limit. Identify every rule this record breaks, and give the v1.0 treatment of Dogwood.

CHECK YOUR UNDERSTANDING

One question before you go

Birch holds 118 shares filled at $72.20, from a 138-share order, with a $70.20 stop and a $76.20 target. In session 5 its bar is O $72.90 / H $76.60 / L $70.00 / C $71.20. Under Atlas-20 v1.0, what is the trade’s net result after the $5.00 round-trip cost?

A. +$467, or +1.98R: the high exceeded the target, so the target exit is recorded.

B. −$241, or −1.02R: both levels lie inside the bar, so the stop is assumed first and 1R = 118 × $2.

C. −$241, or −0.87R: the stop is assumed first, and 1R = 138 ordered shares × $2 = $276.

Sources & further reading

- SEC / Investor.gov · Understanding Order Types ↗
  Market, limit, stop, and stop-limit mechanics; triggered stops do not guarantee execution price. Numerical plans and paths in the chapters are original fictional examples. Access recorded 2026-10-01.
- SEC / Investor.gov · Trading Halts and Delays ↗
  Regulatory versus non-regulatory halts; news-pending halts as the most common regulatory halt; primary-market halts honored by other U.S. markets; exchanges decide. Does not describe reopening prices; Atlas-20 halt accounting is a teaching convention. Access recorded 2026-10-01.
- Bailey, Borwein, López de Prado & Zhu · The Probability of Backtest Overfitting ↗
  Original research on strategy selection and backtest overfitting. Textbook examples do not implement or claim a computed probability of backtest overfitting. Access recorded 2026-10-01.

Teaching examples are fictional unless labelled historical. Inspect this chapter's evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

Evidence before conviction ↗ From chart idea to trade plan ↗
Statistics for a skeptical chart reader ↗ Case studies: follow the decision through ↗
Practice & review this part →